Exercise 12.2

1. Find the arithmetic mean in each of the following:
(i) 4, 6, 10, 12, 15, 20, 25, 28, 30.

X
4
6
10
12
15
20
25
28
30
$\sum X = 150$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{150}{9}$$
$$\boldsymbol{\bar{X} = 16.67}$$

(ii) 12, 18, 19, 0, -19, -18, -12

X
12
18
19
0
-19
-18
-12
$\sum X = 0$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{0}{7}$$
$$\boldsymbol{\bar{X} = 0}$$

(iii) 6.5, 11, 12.3, 9, 8.1, 16, 18, 20.5, 25

X
6.5
11
12.3
9
8.1
16
18
20.5
25
$\sum X = 126.4$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{126.4}{9}$$
$$\boldsymbol{\bar{X} = 14.044}$$

(iv) 8, 10, 12, 14, 16, 20, 22

X
8
10
12
14
16
20

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{102}{7}$$
$$\boldsymbol{\bar{X} = 14.57}$$

22
$\sum X = \mathbf{102}$

2. Following are the heights in (inches) of 12 students. Find the median height.
55, 53, 54, 58, 60, 61, 62, 56, 57, 52, 51, 63.

By arranging

$$51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 62, 63$$

Since number of observations is even i.e. $n = 12$

$$\tilde{X} = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{th} observation + \left(\frac{n+2}{2}\right)^{th} \ observation\right]$$
$$\tilde{X} = \frac{1}{2}\left[\left(\frac{12}{2}\right)^{th} observation + \left(\frac{12+2}{2}\right)^{th} \ observation\right]$$
$$\tilde{X} = \frac{1}{2}[(6)^{th} observation + (7)^{th} \ observation]$$
$$\tilde{X} = \frac{1}{2}[56 + 57]$$
$$\tilde{X} = \frac{113}{2}$$
$$\boldsymbol{\tilde{X} = 56.5 \ inches}$$

3. Following are the earnings (in Rs.) of ten workers: 88, 70, 72, 125, 115, 95, 81, 90, 95, 90.
Calculate (i) Arithmetic Mean

X
88
70
72
125
115
95
81
90
95
90
$\sum X = \mathbf{921}$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{921}{10}$$
$$\boldsymbol{\bar{X} = 92.1}$$

(ii) Median

By arranging

$$70, 72, 81, 88, 90, 90, 95, 95, 115, 125$$

Since number of observations is even i.e. $n = 10$

$$\tilde{X} = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{th} observation + \left(\frac{n+2}{2}\right)^{th} \ observation\right]$$
$$\tilde{X} = \frac{1}{2}\left[\left(\frac{10}{2}\right)^{th} observation + \left(\frac{10+2}{2}\right)^{th} \ observation\right]$$
$$\tilde{X} = \frac{1}{2}[(5)^{th} observation + (6)^{th} \ observation]$$
$$\tilde{X} = \frac{1}{2}[90 + 90]$$
$$\tilde{X} = \frac{180}{2}$$
$$\boldsymbol{\tilde{X} = 90}$$

(iii) Mode

$$Mode = \hat{X} = the \ most \ frequent \ observation$$
$$\hat{X} = 90, 95$$

Note: In the data:

- 90 appears 2 times
- 95 appears 2 times
- Others appear once.

4. The Marks obtained by the students in the subject of English are given below.

Marks obtained	$\mathbf{15-19}$	$\mathbf{20-24}$	$\mathbf{25-29}$	$\mathbf{30-34}$	$\mathbf{35-39}$
Frequency	$\mathbf{9}$	$\mathbf{18}$	$\mathbf{35}$	$\mathbf{17}$	$\mathbf{5}$

Find: (i) Arithmetic mean of their marks by direct and short formula.

Arithmetic mean by direct method:

Class limits	f	X	fX
$15-19$	9	17	153
$20-24$	18	22	396
$25-29$	35	27	945
$30-34$	17	32	544
$35-39$	5	37	185
	$\sum f$ =**84**		$\sum fX = 2223$

$$\bar{X} = \frac{\sum fX}{\sum f}$$

$$\bar{X} = \frac{2223}{84}$$

$$\bar{X} = 26.46$$

Arithmetic mean by short formula: Let $A = 27$

Class limits	f	X	$D = X - A$ $D = X - 27$	fD
$15-19$	9	17	-10	-90
$20-24$	18	22	-5	-90
$25-29$	35	27	0	0
$30-34$	17	32	5	85
$35-39$	5	37	10	50
	$\sum f$ =**84**			$\sum fD = -45$

$$\bar{X} = A + \frac{\sum fD}{\sum f}$$

$$\bar{X} = 27 + \frac{-45}{84}$$

$$\bar{X} = 27 - 0.5357$$

$$\bar{X} = 26.46$$

(ii) Median of their marks.

Class	f	Class Boundaries $(C.B)$	Cumulative Frequency $(c.f)$	
$15-19$	9	$14.5-19.5$	9	
$20-24$	18	$19.5-24.5$	$9+18=27$	
$25-29$	35	$24.5-29.5$	$27+35=62$	$\rightarrow$ ***Median Class***
$30-34$	17	$29.5-34.5$	$62+17=79$	
$35-39$	5	$34.5-39.5$	$79+5=84$	

Here $n = 84$, So $\frac{n}{2} = \frac{84}{2} = 42$

Hence 42^{th} item lie in $24.5-29.5$

$$Median = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$

$$\tilde{X} = 24.5 + \frac{5}{35}\left(\frac{84}{2} - 27\right)$$
$$\tilde{X} = 24.5 + (0.1428)(42 - 27)$$
$$\tilde{X} = 24.5 + (0.1428)(15)$$
$$\tilde{X} = 24.5 + 2.142$$
$$\tilde{X} = 26.642$$

5. Given below is a frequency distribution.

Class Interval	**5 − 9**	**10 − 14**	**15 − 19**	**20 − 24**	**25 − 29**
Frequency	**1**	**8**	**18**	**11**	**2**

Find the mode of the frequency distribution.

Class Interval	Frequency (f)	Class Boundaries$(C.B)$	
5 − 9	1	4.5 − 9.5	
10 − 14	8	9.5 − 14.5	
15 − 19	18	14.5 − 19.5	$\rightarrow$ ***Modal Class***
20 − 24	11	19.5 − 24.5	
25 − 29	2	24.5 − 29.5	

$$Modal\ Class = the\ most\ frequent\ observation$$
$$= 14.5 - 19.5$$

$$Mode = l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h$$
$$\hat{X} = 14.5 + \frac{(18 - 8)}{(18 - 8) + (18 - 11)} \times 5$$
$$\hat{X} = 14.5 + \frac{10}{10 + 7} \times 5$$

$$\hat{X} = 14.5 + \frac{50}{17}$$
$$\hat{X} = 14.5 + 2.94$$
$$\boldsymbol{\hat{X} = 17.47}$$

6. Ten boys work on a petrol pump station. They get weekly wages as follows: Wages (in Rs.) $\mathbf{4250, 4350, 4400, 4250, 4350, 4410, 4500, 4300, 4500, 4390}$. Find the arithmetic mean by short formula, median and mode of their wages.

Arithmetic mean by short formula: Let $A = 4350$

Wages (X)	$D = X - 4350$
4250	−100
4250	−100
4300	−50
4350	0
4350	0
4390	40
4400	50
4410	60
4500	150
4500	150
	$\sum D = 200$

$$\bar{X} = A + \frac{\sum D}{n}$$
$$\bar{X} = 4350 + \frac{200}{10}$$

$$\bar{X} = 4350 + 20$$
$$\boldsymbol{\bar{X} = 4370\ Rs.}$$

Median: By arranging

$$4250, 4250, 4300, 4350, 4350, 4390, 4400, 4410, 4500, 4500$$

Since number of observations is even i.e. $n = 10$

$$\tilde{X} = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{th} observation + \left(\frac{n+2}{2}\right)^{th}\ observation\right]$$
$$\tilde{X} = \frac{1}{2}\left[\left(\frac{10}{2}\right)^{th} observation + \left(\frac{10+2}{2}\right)^{th}\ observation\right]$$
$$\tilde{X} = \frac{1}{2}[(5)^{th} observation + (6)^{th}\ observation]$$
$$\tilde{X} = \frac{1}{2}[4350 + 4390]$$
$$\tilde{X} = \frac{8740}{2}$$
$$\boldsymbol{\tilde{X} = 4370\ Rs.}$$

Mode:

$$Mode = \hat{X} = the\ most\ frequent\ observation$$
$$\boldsymbol{\hat{X} = 4250, 4350\ and\ 4500}$$

Note: From data:

- 4250 occurs 2 times
- 4350 occurs 2 times
- 4500 occurs 2 times
- Others occur once.

7. The arithmetic mean of 45 numbers is 80. Find their sum.

$$\bar{X} = 80$$
$$n = 45$$
$$\sum X = ?$$

By using formula

$$\bar{X} = \frac{\sum X}{n}$$
$$80 = \frac{\sum X}{45}$$
$$80 \times 45 = \sum X$$
$$3600 = \sum X$$
$$\boldsymbol{\sum X = 3600}$$

8. Five numbers are 1, 4, 0, 7, 9. Find their mean, median and mode.

Mean:

X
1
4
0
7
9
$\sum X = 21$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{21}{5}$$
$$\boldsymbol{\bar{X} = 4.2}$$

Median: By arranging

$$0, 1, 4, 7, 9$$

Since number of observations is odd i.e. $n = 5$

$$\tilde{X} = \left(\frac{n+1}{2}\right)^{th} \ observation$$
$$\tilde{X} = \left(\frac{5+1}{2}\right)^{th} \ observation$$
$$\tilde{X} = 3^{rd} \ observation$$
$$\boldsymbol{\tilde{X} = 4}$$

Mode:

$$\hat{X} = the \ most \ frequent \ observation$$
$$\hat{X} = no \ mode \qquad \because no \ repetition$$

9. A set of data contains the values as $148, 145, 160, 157, 156, 160$. Show that $Mode > Median > Mean$.

Mean:

X
148
145
160
157
156
160
$\sum X = 926$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{926}{6}$$
$$\boldsymbol{\bar{X} = 154.33}$$

Median: By arranging

$$145, 148, 156, 157, 160, 160$$

Since number of observations is even i.e. $n = 6$

$$\tilde{X} = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{th} observation + \left(\frac{n+2}{2}\right)^{th} \ observation\right]$$
$$\tilde{X} = \frac{1}{2}\left[\left(\frac{6}{2}\right)^{th} observation + \left(\frac{6+2}{2}\right)^{th} \ observation\right]$$
$$\tilde{X} = \frac{1}{2}[(3)^{rd} observation + (4)^{th} \ observation]$$
$$\tilde{X} = \frac{1}{2}[156 + 157]$$
$$\tilde{X} = \frac{313}{2}$$
$$\boldsymbol{\tilde{X} = 156.5}$$

Mode:

$$\hat{X} = the \ most \ frequent \ observation$$
$$\hat{X} = 160$$

So, $Mode \ (160) > Median \ (156.5) > Mean \ (154.3)$

10. The monthly attendance of 10 students for their lunch in the hostel is recorded as: $21, 15, 16, 18, 14, 17, 15, 12, 13, 11$. Find the median and mode of the attendance. Also find the mean if $D = A - 20$.

Here $D = A - 20$. So $Assumed \ Mean = 20$

Attendance (A)	$D = A - 20$
11	-9
12	-8
13	-7
14	-6

15	-5
15	-5
16	-4
17	-3
18	-2
21	$+1$
	$\sum \boldsymbol{D} = \mathbf{-48}$

$$\bar{X} = A + \frac{\sum D}{n}$$
$$\bar{X} = 20 + \frac{-48}{10}$$
$$\bar{X} = 20 - 4.8$$
$$\boldsymbol{\bar{X} = 15.2}$$

11. On a prize distribution day, 50 students brought pocket money as under:

Rupees	**5 − 10**	**10 − 15**	**15 − 20**	**20 − 25**	**25 − 30**
Frequency (f)	**12**	**9**	**18**	**7**	**4**

(i) Find the median and mode of the above data.

Median:

Class	$\boldsymbol{f}$	**Class Boundaries** $(\boldsymbol{C.B})$	**Cumulative Frequency** $(\boldsymbol{c.f})$	
5 − 10	12	5 − 10	12	
10 − 15	9	10 − 15	21	
15 − 20	18	15 − 20	39	$\rightarrow \boldsymbol{Median\ Class}$
20 − 25	7	20 − 25	46	
25 − 30	4	25 − 30	50	

Here $n = 50$, So $\frac{n}{2} = \frac{50}{2} = 25$

Hence 25^{th} item lie in $15 - 20$

$$Median = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$
$$\tilde{X} = 15 + \frac{5}{18}\left(\frac{50}{2} - 21\right)$$
$$\tilde{X} = 15 + (0.2778)(25 - 21)$$
$$\tilde{X} = 15 + (0.2778)(4)$$
$$\tilde{X} = 15 + 1.1112$$
$$\boldsymbol{\tilde{X} = 16.1112}$$

Note: In grouped data, if classes are continuous (e.g. $5 - 10$, $10 - 15$), class width h is taken directly, but if classes are inclusive (e.g. $15 - 19$, $20 - 24$), we first use class boundaries to remove gaps, and then find h.

Mode:

Class Interval	**Frequency** $(\boldsymbol{f})$	**Class Boundaries**$(\boldsymbol{C.B})$	
5 − 10	12	5 − 10	
10 − 15	9	10 − 15	
15 − 20	18	15 − 20	$\rightarrow Modal\ Class$
20 − 25	7	20 − 25	
25 − 30	4	25 − 30	

$$Modal\ Class = the\ most\ frequent\ observation$$
$$= 15 - 20$$

$$Mode = l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h$$
$$\hat{X} = 15 + \frac{(18-9)}{(18-9)+(18-7)} \times 5$$
$$\hat{X} = 15 + \frac{9}{9+11} \times 5$$
$$\hat{X} = 15 + \frac{45}{20}$$
$$\hat{X} = 15 + 2.25$$
$$\boldsymbol{\hat{X} = 17.25}$$

(ii) Find the arithmetic mean of the data given above using coding method.

Let $A = 17.5$, class width $h = 5$

Class	f	Midpoint (X)	$U = \frac{X-A}{h}$, $U = \frac{X-17.5}{5}$	fU
$5 - 10$	12	7.5	-2	-24
$10 - 15$	9	12.5	-1	-9
$15 - 20$	18	17.5	0	0
$20 - 25$	7	22.5	1	7
$25 - 30$	4	27.5	2	8
	$\sum f = 50$			$\sum fU = -18$

$$\bar{X} = A + \frac{\sum fU}{\sum f} \times h$$

$$\bar{X} = 17.5 + \frac{-18}{50} \times 5$$
$$\bar{X} = 17.5 - \frac{90}{50}$$
$$\bar{X} = 17.5 - 1.5$$
$$\boldsymbol{\bar{X} = 15.7}$$

12. The arithmetic mean of the ages of 20 *boys* is 13 *years*, 4 *months* and 5 *days*. Find the sum of their ages. If one of the boys is of age exactly 15 years. What is the average age of the remaining boys?

$$No.of\ boys = n = 20$$
$$Arithmetic\ mean = \bar{X} = 13\ years, 4\ months\ and\ 5\ day$$
$$\bar{X} = 13\ years + 4\ months + 5\ day$$
$$\bar{X} = \left(13 + \frac{4}{12} + \frac{5}{365}\right)\ years$$
$$\bar{X} = (13 + 0.3333 + 0.0137)\ years$$
$$\bar{X} = 13.374\ years$$
$$Sum\ of\ ages = \sum X = ?$$
$$Average\ age\ of\ the\ remaining\ boys = ? \qquad (When\ age\ of\ one\ boy = 15\ years)$$

By using formula

$$\bar{X} = \frac{\sum X}{n}$$
$$13.374 = \frac{\sum X}{20}$$
$$13.374 \times 20 = \sum X$$
$$266.94 = \sum X$$
$$\boldsymbol{\sum X = 266.94\ years}$$

$$\sum X = 266\ yrs. + 0.94\ yrs.$$
$$\sum X = 266\ yrs. + (0.94\ \times 12)\ months$$
$$\sum X = 266\ yrs. + 11.28\ months$$
$$\sum X = 266\ yrs. + 11\ months + 0.28\ months$$
$$\sum X = 266\ yrs. + 11\ months + (0.28 \times 31)\ days$$
$$\sum X = 266\ yrs. + 11\ months + 8.68\ days$$
$$\boldsymbol{\sum X = 266\ years, 11\ months\ and\ 9\ days}$$

Now

$$Age\ of\ ramining\ 19\ boys = 266.97 - 15$$
$$\boldsymbol{\sum X = 251.94}$$

Now again by using formula

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{251.94}{19}$$
$$\bar{X} = 13.26\ years$$
$$\bar{X} = 13\ yrs. + 0.26\ yrs.$$
$$\bar{X} = 13\ yrs. + (0.26\ \times 12)\ months$$
$$\bar{X} = 13\ yrs. + 3.12\ months$$
$$\bar{X} = 13\ yrs. + 3\ months + 0.12\ months$$
$$\bar{X} = 13\ yrs. + 3\ months + (0.12 \times 30)\ days$$
$$\bar{X} = 13\ yrs. + 3\ months + 3.6\ days$$
$$\boldsymbol{\bar{X} = 13\ years, 3\ months\ and\ 4\ days}$$

13. Calculate the arithmetic mean from the following information:

(i) If $\boldsymbol{D = X - 140}$, $\boldsymbol{\sum D = 500}$ and $\boldsymbol{n = 10}$

Give that

$$D = X - 140$$
$$\sum D = 500$$
$$n = 10$$
$$\bar{X} = ?$$

As $D = X - 140$. By comparing with $D = X - A$, $\Longrightarrow A = 140$

By using formula

$$\bar{X} = A + \frac{\sum D}{n}$$
$$\bar{X} = 140 + \frac{500}{10}$$
$$\bar{X} = 140 + 50$$
$$\boldsymbol{\bar{X} = 90}$$

(ii) If $\boldsymbol{U = \frac{X-130}{6}}$, $\boldsymbol{\sum U = -150}$ and $\boldsymbol{n = 15}$

Give that

$$U = \frac{X - 130}{6}$$
$$\sum U = -150$$
$$n = 15$$
$$\bar{X} = ?$$

As $U = \frac{X-130}{6}$. By comparing with $U = \frac{X-A}{h}$, $\Longrightarrow A = 130$ and $h = 6$

By using formula

$$\bar{X} = A + \frac{\sum U}{n} \times h$$
$$\bar{X} = 130 + \frac{-150}{15} \times 6$$
$$\bar{X} = 130 - \frac{900}{15}$$
$$\bar{X} = 130 - 60$$
$$\boldsymbol{\bar{X} = 70}$$

(iii) If $D = X - 25, \sum fD = 300$ and $\sum f = 20$

Give that

$$D = X - 25$$
$$\sum fD = 300$$
$$\sum f = 20$$
$$\bar{X} = ?$$

As $D = X - 25$. By comparing with $D = X - A, \implies A = 25$

By using formula

$$\bar{X} = A + \frac{\sum fD}{\sum f}$$
$$\bar{X} = 25 + \frac{300}{20}$$
$$\bar{X} = 25 + 15$$
$$\boldsymbol{\bar{X} = 40}$$

(iv) If $U = \frac{X-120}{5}, \sum fU = 60$ and $\sum f = 100$

Give that

$$U = \frac{X - 120}{5}$$
$$\sum fU = 60$$
$$\sum f = 100$$
$$\bar{X} = ?$$

As $U = \frac{X-120}{5}$. By comparing with $U = \frac{X-A}{h}, \implies A = 120$ and $h = 5$

By using formula

$$\bar{X} = A + \frac{\sum fU}{\sum f} \times h$$
$$\bar{X} = 120 + \frac{60}{100} \times 5$$
$$\bar{X} = 120 + \frac{300}{100}$$
$$\bar{X} = 120 + 3$$
$$\boldsymbol{\bar{X} = 123}$$

14. The three children Haris, Maham and Minal made the following scores in a game conducted by a group of teachers in the school.

Haris scores	**50**	**55**	**70**	**85**	**90**
Maham scores	**75**	**60**	**60**	**45**	**53**
Minal scores	**80**	**77**	**66**	**42**	**48**

It is decided that the candidate who gets the highest average score will be awarded rupees 1000. Who will get the awarded amount?

Haris (X_H)	**Maham (X_M)**	**Minal (X_N)**
50	75	80

55	60	77
70	60	66
85	45	42
90	53	48
$\boldsymbol{\sum X_H = 350}$	$\boldsymbol{\sum X_M = 293}$	$\boldsymbol{\sum X_N = 313}$

$$Average\ score\ of\ Haris = \overline{X_H} = \frac{\sum X_H}{n}$$
$$\overline{X_H} = \frac{350}{5}$$
$$\boldsymbol{\overline{X_H} = 70}$$

$$Average\ score\ of\ Maham = \overline{X_M} = \frac{\sum X_M}{n}$$
$$\overline{X_M} = \frac{293}{5}$$
$$\boldsymbol{\overline{X_M} = 58.6}$$

$$Average\ score\ of\ Minal = \overline{X_N} = \frac{\sum X_N}{n}$$
$$\overline{X_N} = \frac{313}{5}$$
$$\boldsymbol{\overline{X_N} = 62.6}$$

Since Haris has the highest average score ($\boldsymbol{70}$). He will get the awarded amount of $\boldsymbol{Rs.\ 1000}$.

15. Given below is a frequency distribution derived by making a substitution as $\boldsymbol{D = X - 20}$.

$\boldsymbol{D}$	$\boldsymbol{-6}$	$\boldsymbol{-4}$	$\boldsymbol{-2}$	$\boldsymbol{0}$	$\boldsymbol{2}$	$\boldsymbol{4}$	$\boldsymbol{6}$
$\boldsymbol{f}$	**1**	**3**	**6**	**20**	**26**	**12**	**2**

Calculate the arithmetic mean.

As $D = X - 20$. By comparing with $D = X - A, \Longrightarrow A = 20$

$\boldsymbol{D}$	$\boldsymbol{f}$	$\boldsymbol{fD}$
-6	1	-6
-4	3	-12
-2	6	-12
0	20	0
2	26	52
4	12	48
6	2	12
	$\boldsymbol{\sum f = 70}$	$\boldsymbol{\sum fD = 82}$

By using formula

$$\bar{X} = A + \frac{\sum fD}{\sum f}$$
$$\bar{X} = 20 + \frac{82}{70}$$
$$\bar{X} = 20 + 1.17$$
$$\boldsymbol{\bar{X} = 21.17}$$

16. Being partners Hafsa and Fatima took part in a quiz programme. They made the following number of points $\boldsymbol{45, 51, 58, 61, 74.48, 46}$ and $\boldsymbol{50}$. Compute the average number of points using deviation $\boldsymbol{D = X - 58}$.

As $D = X - 58$.

By comparing with $D = X - A, \Longrightarrow A = 58$

X	$D = X - 58$
45	-13
51	-7
58	0
61	3
74	16
48	-10
46	-12
50	-8
	$\sum D = -31$

By using formula

$$\bar{X} = A + \frac{\sum D}{n}$$
$$\bar{X} = 58 + \frac{-31}{8}$$
$$\bar{X} = 58 - 3.875$$
$$\bar{X} = 54.125$$

17. A person purchased the following food items:

Food item	Quantity (in Kg)	Cost per Kg (in Rs.)
Rice	**10**	**96**
Flour	**12**	**48**
Ghee	**4**	**190**
Sugar	**3**	**49**
Mutton	**2**	**650**

What is the weighted mean of cost of food items per kg?

Food item	Quantity (W)	Cost per Kg (X)	WX
Rice	10	96	960
Flour	12	48	576
Ghee	4	190	760
Sugar	3	49	147
Mutton	2	650	1300
	$\sum W = 31$		$\sum WX = 3743$

$$\bar{X}_w = \frac{\sum WX}{\sum W}$$
$$\bar{X}_w = \frac{3743}{31}$$
$$\bar{X}_w = 120.74$$

18. For the following data,

Item	Quantity	Cost of item (in thousands)
Washing Machine	**5**	**35**
Heater	**3**	**5**
Stove	**2**	**13**
Dispenser	**6**	**18**

find the weighted mean.

Item	Quantity (W)	Cost in thousands (X)	WX
Washing Machine	5	35	175
Heater	3	5	15
Stove	2	13	26

Dispenser	6	18	108
	$\sum W = 16$		$\sum WX = 324$

$$\bar{X}_w = \frac{\sum WX}{\sum W}$$
$$\bar{X}_w = \frac{324}{16}$$
$$\bar{\boldsymbol{X}}_{\boldsymbol{w}} = \mathbf{20.25}\ \boldsymbol{thousands}$$

19. A company is planning its next year marketing budget across five years: yearly budgets (in million) are: 5, 7, 8, 6, 7. Find the average budget for the next year.

Budget (X)
5
7
8
6
7
$\sum X = 33$

$$\bar{X} = \frac{\sum X}{n}$$
$$\bar{X} = \frac{33}{5}$$
$$\bar{\boldsymbol{X}} = \mathbf{6.6}\ \boldsymbol{million}$$

20. Ahmad obtained the following marks in a certain examination.

Urdu	English	Science	Math	Islamiyat	Computer
78	**65**	**80**	**90**	**85**	**72**

Find the weighted mean if weights 5, 4, 2, 3, 2, 4 respectively are allotted to the subjects.

Subject	Weight (W)	Marks (X)	WX
Urdu	5	78	390
English	4	65	260
Science	2	80	160
Math	3	90	270
Islamiyat	2	85	170
Computer	4	72	288
	$\sum W = 20$		$\sum WX = 1538$

$$\bar{X}_w = \frac{\sum WX}{\sum W}$$
$$\bar{X}_w = \frac{1539}{20}$$
$$\bar{\boldsymbol{X}}_{\boldsymbol{w}} = \mathbf{76.9}$$